

identical to the original file in all aspects. If you take a spectrum analysis of the original file and compare it with the compressed file, the resultant spectrum will be one single line with all the peaks and valleys and spikes matching each other to the 'T'. If not, it means there is some loss that is taking place during compression.

Encoding music using a lossless codec takes a lot of time and does not compress music a great deal. A 10MB file will be compressed to a maximum of about 7MB using WMA (Windows Media Audio) 9. There are other codecs such as AAC (Advance Audio Codec) and Ogg Vorbis, which we will talk about in a little while. But before we talk about these codecs, let's talk about the MP3 codec, the one piece of software that started it all.

An apt example of a lossy codec is MP3. The MP3 codec uses the "perceptual coding" technique. The Fraunhofer institute in Germany, along with Prof. Dieter Seitzer of the University of Erlangen, developed an algorithm that was standardised as the ISO-MPEG Audio Layer-3.

MP3 is far more popular than any other format or codec as it allows for exceptionally small file sizes without much difference in the rendered audio quality. However, there is a large difference statistically between the original uncompressed music file and the resultant MP3 file. This is where perceptual coding steps in.

The trick that "perceptual coding" uses, is it removes information from a particular audio or video file. Users cannot "perceive" the loss of higher or lower frequencies that are present in the original file but are cut off in the compressed MP3 file. The overall music quality does not suffer but the file size reduces due to such loss of information and hence MP3 is called a "lossy codec".

If you run a spectrum analyser on a compressed MP3 file and compare it with the original uncompressed recording, you will find that there is no "one-to-one" match of the two spectrums. Very high or very low frequencies which the codec thinks is